

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

AU-7004(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Robotics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What are the design consideration of grippers?
b) Write down the application of different types of drives used in robot.
2. a) Distinguish between VAL and platy robot programming languages.
b) Write down the application of different types of drives used in robot.
3. a) With the help of neat sketch discuss Robot anatomy.
b) Discuss various Robot part and write their functions.
4. a) Discuss industrial application of vision control Robot.
b) Define the following sensors.
 - i) Pizo electric
 - ii) L.V.D.T.
5. a) The coordinates of point P in frame $[3.0 \ 2.0 \ 1.0]^T$. The position vector P is rotated about the Z-axis by 45° . Find the coordinates of point Q the new position of point P.
b) Write down the steps used in Robot programming language-VAL.

AU-7004(2)-CBGS

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[2]

6.
 - a) Why Robots are needed? Enlist its applications.
 - b) What is Work envelope? How Robots are classified?

7.
 - a) Briefly discuss the mode of programming industrial robots from point of view of safety.
 - b) Define the term “Encoder” and “Compliance”.

8. Write brief notes on followings:
 - a) Implementation of Robots in industries
 - b) Safety consideration for robot operations
